An open-data study on wearable stress detection (WESAD)

Stress detection that still works when the schedule changes

How a benchmark can be won by reading the clock, a simple model that reads the body instead, and what physiological coupling graphs reveal about stress.

Salman Almuhammad Alali, November 2026

The study at a glance

Stress detection that still works when the schedule changes

Open WESAD dataset: 15 people, chest and wrist wearables, a lab stress test. Every number below comes from that data.

1 The data

Chest + wrist sensors
15 people

Two orders of events:

Order A, 8 people

Rest › Fun › Calm › Stress › Calm

Order B, 7 people

Rest › Stress › Calm › Fun › Calm

Stress test early or late in the session

2 A hidden shortcut

Minutes since the start, nothing else:

95.7%

accuracy on new people

Original paper	93.1%
Clock only	95.7%

3 A fair test

Train on order A

Test on order B, then reverse

Drop 4 measures that drift with time

A simple model plus a body-signal coupling graph

4 The result



Heart stops following movement under stress
15 of 15 people

For VR training: works on a new trainee · interpretable · near real time · body + behaviour

Why this matters for VR training

A VR training system should notice when a trainee is stressed: for a person it has never seen, while the scenario is running, and in a way an instructor can understand.



Across people

Works on a new trainee without retraining



Interpretable

Built on named body measures an instructor can read



In real time

Updates within seconds, using only the past

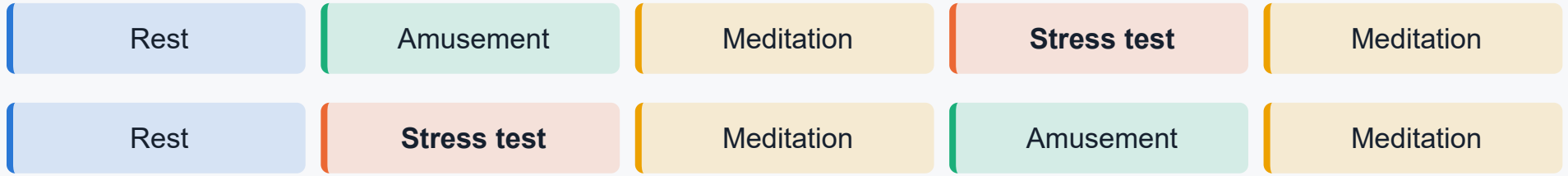
The data: WESAD

15 people
in a lab, about 100 minutes each, wearing both sensors at once

Chest sensor
heart (ECG), breathing, sweat, muscle, temperature, movement

Wrist sensor
pulse, sweat, temperature and movement, like a smartwatch

Two orders of events

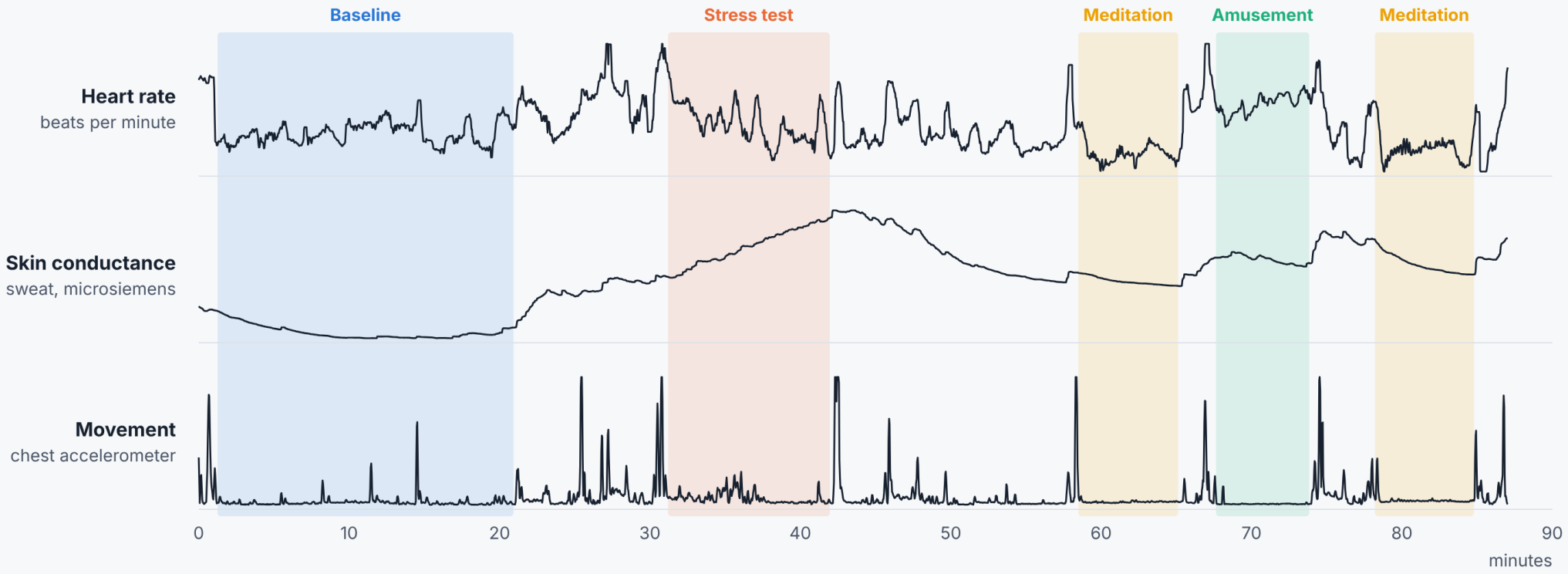


Rest about 20 min, amusement (funny videos) about 6 min, stress test about 10 min (a public speech and mental arithmetic), meditation about 7 min. The task: tell stress from non-stress in each minute.

Data: Schmidt et al., “Introducing WESAD”, ICMI 2018. Open for scientific, non-commercial use.

What stress does to the body

One typical participant’s whole session. In the stress test, heart rate and sweating rise. Skin conductance also creeps up all session long: a slow drift that tracks time.



Testing fairly: never on someone the model has seen



Train on 14 people, test on the 15th, and repeat so every person is tested once.

Nothing leaks

Every setting is learned from the training people only.

Scored per person

Each person gets a score; the result is their average.

Plan fixed first

Each analysis was written down in git before it was run.

The trap: a clock beats the benchmark



Rest
Amusement
Stress test
Meditation

95.7%

accuracy from the minutes since the start alone

The original paper reached 93.1% with physiology. The timetable is almost the same for everyone, so time reveals the condition.

The fix: test on a schedule the model never saw

Train: 8 people, order A



Test: 7 other people, order B



Then the reverse. Order A did amusement before the stress test; order B after it.



Per-person macro-F1 for stress vs non-stress, mean of 15 people. * Scored without the first two baseline minutes.

The model in five simple steps



1. Cut
One-minute windows, each inside one condition

2. Describe
38 body measures + a graph of how signals move together

3. Remove the clock
Drop 4 measures that drift with time (temperature, sweat level)

4. Classify
Simple, readable model: p = σ(w·z + b)

5. Test fairly
New people and a new schedule, never seen in training

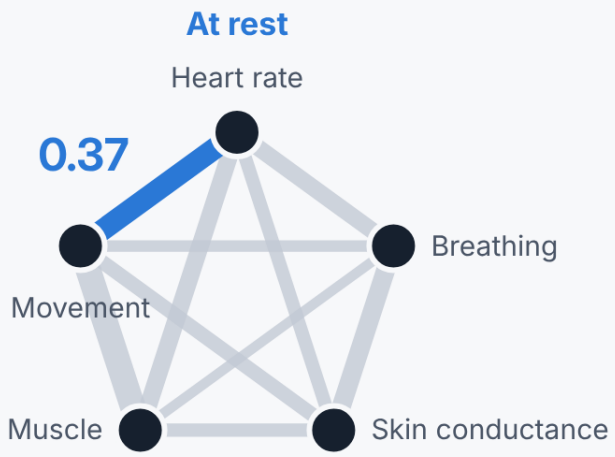
In plain words

The model weighs each body measure, adds them up, and turns the sum into a probability of stress.

$p = \sigma(w \cdot z + b)$

z: body measures, standardised on the training people. w, b: learned weights. σ squashes any number into 0 to 1.

Coupling graphs: how body signals move together



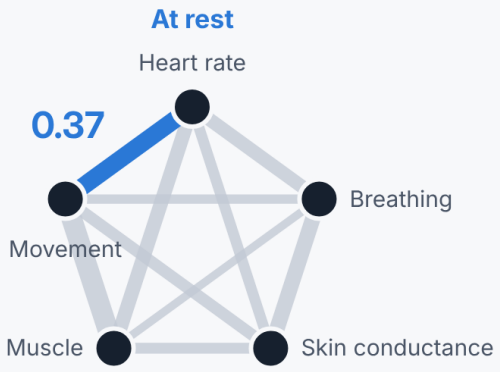
Each signal is a node. An edge says how strongly two signals move together within one minute.

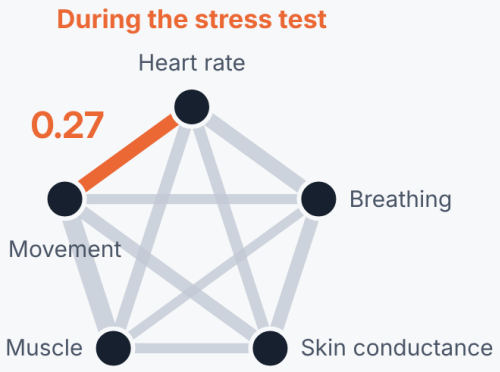
The edges are learned with a random-walk model from the author's published work: influence spreads along the graph like heat, and the weights that best explain the observed links are kept.

$$\min \|\tilde{T} - \alpha \cdot \exp(-L(w))\|^2 + \lambda \cdot \Sigma w,\ w \geq 0$$

It separates direct links from indirect ones that pass through a third signal.

Under stress, the heart stops following movement







15 of 15

people show the drop

Stress only

the drop does not appear under amusement

Robust

to matching on how much people moved

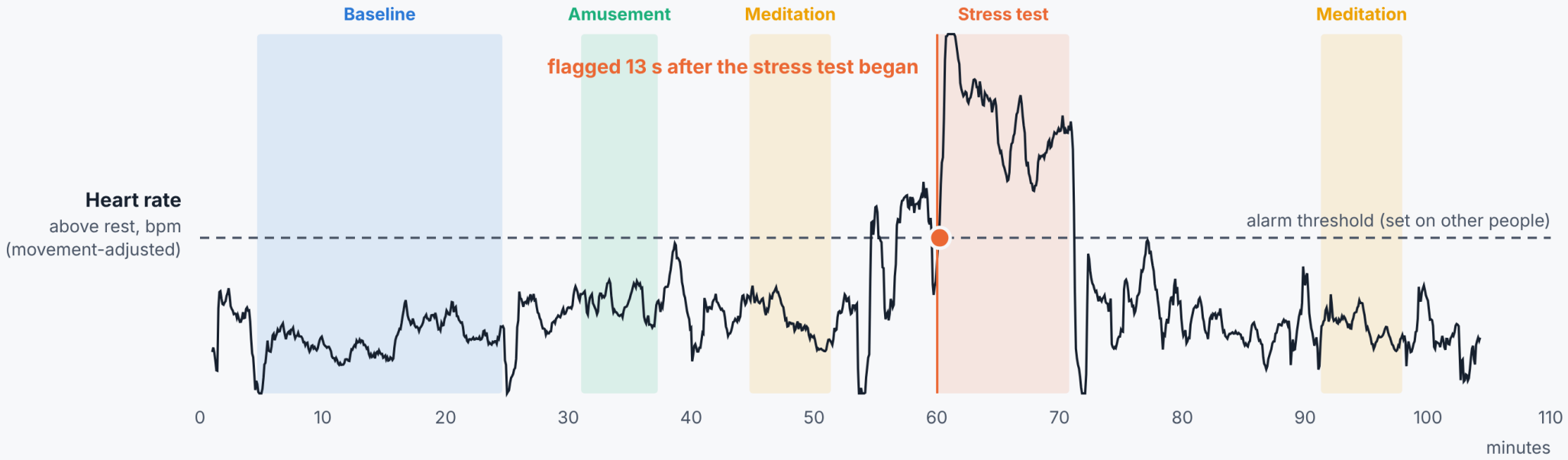
Where the results stand

Model	New people	New people and new schedule
Original WESAD paper (2018)	93.1% accuracy	not reported
Best published, same test	about 94–95% accuracy	not reported
Clock only	95.7% accuracy	66.9% accuracy
This study	94.0% accuracy	0.943 F1 per person

Some reports of 97–99% let the same person appear in training and testing, or tune on the test set; a 2026 survey found such leakage in 4 of 22 studies.

Sources: Schmidt et al., ICMI 2018; Bobade & Vani 2020; SELF-CARE; EDA survey, arXiv 2609.22622 (2026).

A real-time version



5 s
between updates, using only the past

0.2 s
to compute each update of the graph

13 s
median time to flag the stress test

What carries over to stress estimation in VR

The study was designed around what a VR training system needs from a stress model.

Works on a new trainee

Tested on people and on a schedule it never saw

Interpretable

Named body measures and a readable coupling marker

Near real time

Updates every 5 s; stress onset flagged in a median of 13 s

Body and behaviour

Links body signals to movement; ready for head, hand and gaze

Reproducible

Plan fixed before each analysis; one script reruns everything

VR scenarios also run in fixed orders, so the clock shortcut and the cross-order test apply to them directly.

Limitations and what I learned

Limitations

- 15 people, one lab, one kind of stressor
- Speaking and moving come with the stress test
- Graph features confirmed the physiology but did not add significant accuracy with 15 people
- Signal cleaning used whole recordings, offline

Lessons

- Test on a new schedule, not only new people
- Watch for measures that drift with time in session
- Judge movement-aware measures where movement varies on its own
- Fix the analysis plan before running it

Every result, including the ones that did not work, is reported in the repository.

Next steps in VR

- Add head, hand and gaze signals as nodes of the coupling graph
- Counterbalance or randomise scenario order, and always report a context-only baseline
- Use the causal index to adapt a scenario while it runs
- Confirm on a second dataset with more people and other stressors

Code, full report and every result: github.com/SAamal1076/wesad-stress-graphs

Salman Almuhammad Alali. Code written with an AI assistant (Claude) under the author’s direction.